Finance Research Letters 86 (2025) 108518

Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl

When nature disrupts: biodiversity risk and corporate supply chain resilience

Wenwen Jin

School of Business, Wenzhou University, Wenzhou, 325035, Zhejiang, China

ARTICLE INFO

JEL:
Q56
L14
G32

Keywords:
Biodiversity risk
Supply chain resilience
Ecological risk

ABSTRACT

Although climate change risks have been extensively examined, biodiversity risk has emerged as a pressing yet understudied ecological threat to supply chain resilience (SCR). Motivated by the increasing disruptions from biodiversity degradation, this study explores how biodiversity risk shapes SCR in Chinese listed firms. Using fixed effects and instrumental variable approaches, we find that biodiversity risk significantly undermines SCR. The negative effects are especially pronounced in firms with limited diversification, fewer female directors, a manufacturing focus, and non-state ownership. Mechanism tests reveal that maturity mismatch and agency costs channel the negative impact. These findings highlight biodiversity risk as a distinct source of supply chain vulnerability and offer insights into enhancing SCR.

1. Introduction

Natural Resource-Based View (NRBV) holds that competitive advantage comes not only from traditional economic and technological resources but also from strategic capabilities developed under ecological constraints (Hart, 1995). Biodiversity loss exemplifies such constraints, as ecosystem degradation and species decline erode the natural capital on which supply chains rely, thereby weakening their resilience (Polasky and Daily, 2021; Beck-O'Brien and Bringezu, 2021). Although substantial enterprise value is exposed to biodiversity risk, disclosure remains limited, with downstream vulnerabilities often overlooked (Lähtinen et al., 2016; Carvalho et al., 2023).

Extensive research has investigated how climate change disrupts SCR (Ali et al., 2023; Andreoni and Miola, 2015; Ghadge et al., 2020), leaving biodiversity risk underexplored. Since climate and biodiversity risks differ in temporal dynamics, sectoral exposure, and market valuation (Giglio et al., 2023), there is an urgent need to treat biodiversity as an independent source of risk.

Building on this gap, this study seeks to answer three questions: Does biodiversity risk represent a critical environmental constraint undermining SCR? Through which mechanisms does it affect SCR? Do firm characteristics such as diversification, governance structure, industry attributes, and ownership type alter its impact?

This study finds that biodiversity risk significantly undermines SCR. It extends the NRBV by incorporating biodiversity as a distinct source of risk, thereby broadening the framework's application beyond climate risks. It also adds to limited research on corporate responses to biodiversity risks (Ahmad and Karpuz, 2024; Li et al., 2025). By revealing maturity mismatch and agency costs as mechanisms, this study offers insights into firms' adaptive capacity under environmental uncertainty. Finally, by focusing on China, it underscores the policy relevance of integrating biodiversity into corporate strategy, addressing how firms can sustain SCR under ecological constraints and offering implications for emerging economies.

E-mail address: 1594181124@qq.com.

https://doi.org/10.1016/j.frl.2025.108518
Received 12 March 2025; Received in revised form 18 September 2025; Accepted 20 September 2025
Available online 21 September 2025

The remainder of the paper is structured as follows. Section 2 reviews the theoretical framework, Section 3 describes the methodology, Section 4 presents the results, Section 5 discusses the findings, and Section 6 concludes.

2. Theories

Existing research has identified three main categories of factors influencing SCR: advanced technology, internal capacity, and external environmental conditions. Advanced technologies such as AI, blockchain, digital transformation, and digital finance enhance efficiency and adaptability (Wang et al., 2024b; Gupta et al., 2023; Ma et al., 2025; Ge & Bao, 2024; Qi et al., 2024; Li et al., 2024; Pandey et al., 2024). Internal factors, including asset specificity, human capital, leadership, institutional involvement, and network design, also play a critical role (Tennakoon et al., 2025; Zhou et al., 2025; Zhong et al., 2024; Ahmed et al., 2025; Wang et al., 2024a; Türkeş et al., 2024). Meanwhile, external constraints such as limited market access and regulatory gaps pose major barriers to resilience (Liu et al., 2024; Erol et al., 2024). Evidence from the UK and China further highlights the importance of connectivity and access to resources in shaping resilience outcomes (Brandon-Jones, 2014; Liu et al., 2025).

Table 1

Extending beyond traditional resources, the NRBV emphasizes that ecological factors, including biodiversity, are crucial for sustaining competitive advantage (Hart, 1995). As a form of natural capital, biodiversity ensures resource security and production continuity. Its degradation disrupts supply chains and reshapes sustainability strategies (Yin et al., 2025; Fu et al., 2025), potentially limiting firms' adaptive capacity. It also reduces future cash flows and weakens firm performance (Huang et al., 2024; Bach et al., 2025), which is likely to undermine the capacity of supply chains to recover after disruptions, thus posing challenges to SCR.

Additionally, biodiversity risk influences corporate disclosure and transparency (He et al., 2025) but remains mispriced. Because biodiversity-related financial risks often materialize earlier than climate risks (Giglio et al., 2023; Kedward et al., 2023), they generate heightened uncertainty. According to real options theory (Bulan, 2005), such uncertainty combined with the irreversibility of biodiversity loss discourages firms from committing to long-term green strategies (Trinh, 2023; Li et al., 2025). Reluctance to pursue sustainable initiatives restricts the development of resilient capabilities and weakens the foundations of SCR.

Thus, we propose the following hypothesis:

H1: Biodiversity risk weakens corporate SCR.

3. Methodology

3.1. Sample and data sources

To investigate the impact of biodiversity risk on corporate SCR, this study adopts a firm-level panel data approach. It examines

Table 1
Some existing studies on factors influencing corporate SCR.

Article	Factors	Conclusions
Wang et al. (2024b)	Artificial Intelligence (AI)	AI boosts resilience in the primary sector by improving information sharing and response efficiency.
Gupta et al. (2023)	AI, blockchain technology	Blockchain is more effective than AI in strengthening supply chain financial resilience, especially under high environmental dynamism.
Ma et al. (2025)	Intelligent development	Intelligent development enhances SCR by improving efficiency, boosting productivity, and mitigating the long-whip effect.
Ge & Bao (2024)	Digital transformation	Digital transformation optimizes supply-demand alignment, stabilizes connections, fosters supplier innovation, and strengthens SCR.
Qi et al. (2024)	Digital transformation	Digitalization markedly strengthens SCR.
Li et al. (2024)	Digital finance	Digital finance enhances SCR.
Pandey et al. (2024)	Blockchain technology	The study uncovers 21 blockchain-related factors supporting SCR and sustainability.
Tennakoon et al. (2025)	Transaction cost determinants and governanc	Asset specificity, transaction frequency, and transaction uncertainty were closely linked to the constructs of SCR.
Zhou et al. (2025)	Innovative human capital	Supply chain resilience benefits from innovative human capital.
Zhong et al. (2024)	Organizational factors	Supply chain leadership is essential for advancing agri-food SCR.
Ahmed et al.(2025)	Organizational factors	Examining how environmental, social, and governance (ESG) dimensions interact with soft organizational factors is crucial for strengthening supply chain risk mitigation and resilience in large projects.
Wang et al. (2024a)	Organizational factors	Party organization participation in governance shows an inverted U-shaped effect on supply chain stability.
Türkeş et al. (2024)	Proactive, reactive, network design capabilities	These capabilities jointly explain 56.9% of the variance in food SCR.
Liu et al. (2024)	Market accessibility	Market accessibility enhances SCR by improving efficiency and adaptability.
Erol et al. (2024)	Regulations, internal governance	Weak regulations, insufficient incentives, and ineffective governance and supply chain policies are key barriers to SCR.
Brandon-Jones et al. (2014)	Supply chain connectivity and information sharing resources	Supply chain connectivity and information sharing contribute to developing visibility capabilities, which in turn strengthen both resilience and robustness.
Liu et al. (2025)	Resource, production capacity constraints	Cobalt reserves, workforce, cobalt self-sufficiency rate, and recycling technology are major barriers to SCR.

Chinese A-share listed firms from 2003 to 2023, a period encompassing key environmental regulatory changes in China, which allows analysis of the dynamic effects of biodiversity risk. The SCR index is constructed using financial data from the Wind and the CNRDS databases, while biodiversity risk data are obtained from He et al. (2024). To ensure consistency and reliability, we exclude ST/PT firms and observations with missing values for key variables. Continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers, resulting in a final sample of 47,107 firm-year observations.

3.2. Variable selection

3.2.1. Dependent variable

Drawing from Qi et al. (2024), Ma et al. (2025), and Lu et al. (2024) with further refinements, this study measures SCR using the entropy weight method based on four sub-indicators: resistance, stability, recovery, and growth. Resistance reflects a firm's ability to withstand disruptions, calculated as the log ratio of receivables and prepayments to revenue. Stability indicates operational consistency, measured by the proportion of transaction amounts with the top five customers. Recovery represents post-shock restoration capacity, proxied by the deviation between production and demand. Growth captures a firm's potential to rebound and improve, measured by the operating revenue growth.

3.2.2. Independent variable

This study measures corporate biodiversity risk using the dataset from He et al. (2024), based on word frequency in annual reports. Biodiversity risk is identified through text mining, with a binary indicator assigned: taking the value 1 if biodiversity-related terms appear more than twice, and 0 otherwise.

3.2.3. Control variable

Building on related literature (Su et al., 2025; Lyu, 2025), this study includes a set of commonly used control variables: firm size, market value, leverage ratio, and liquidity. In addition, asset tangibility and ownership concentration are incorporated to enhance robustness. Detailed variable descriptions are provided in Table 2, and summary statistics are presented in Table 3.

3.3. Model

To explore the impact of biodiversity risk on firms' SCR, this study employs a fixed effects panel model. This methodology is important because it accounts for firm-specific traits while also capturing industry- and year-specific shocks, thereby ensuring credible identification of the effect of biodiversity risk on SCR.

$$Res = \alpha + \beta_1 BioRisk + \beta_2 Controls + year + firm + industry + \epsilon \quad (1)$$

Res denotes SCR for firm i in year t, *BioRisk* measures biodiversity risk exposure, scaled by 10 for analytical convenience. Controls

Table 2
Variable description.

Variable type	Variable Symbol	Description	Measurement
Dependent variable	Res	Supply Chain Resilience	Measured using the entropy weight method, combining four indicators to construct the SCR index.
Independent variable	BioRisk	Biodiversity Risk	Derived from the frequency of biodiversity-related terms, calculated using text mining of Chinese listed firms' annual reports. Equals 1 if biodiversity-related terms appear more than twice, and 0 otherwise.
	Bioconcern1	Biodiversity Concern	Ratio of biodiversity-related term frequency to the total word frequency in the annual report.
	Bioconcern2	Biodiversity Concern	Ratio of biodiversity-related Chinese characters to the total number of Chinese characters in the annual report.
	Bioconcern3	Biodiversity Concern	Ratio of biodiversity-related characters to the total number of characters in the annual report (including punctuation marks and numbers).
	Ressub1	Supply chain resistance	Capital occupancy by upstream and downstream enterprises in the supply chain.
	Ressub2	Supply chain stability	The share of transaction amounts with the top five customers.
	Ressub3	Supply Chain Recovery	Deviation of production fluctuations from demand fluctuations.
	Ressub4	Supply Chain Growth Ability	Operating growth generated by upstream and downstream supply chain enterprises.
Control variable	size	Firm size	Logarithm of the firm's total assets.
	TobinQ	Market Valuation Ratio	The ratio of total market value to total asset.
	lev	Leverage	Debt-to-asset ratio.
	cash	Cash flow	Current net cash flow divided by total year-end assets.
	tangibility	Tangible Asset Ratio	The ratio of tangible assets to total assets.
	topten	Ownership concentration	Ownership share of the top ten shareholders.

Table 3
Summary statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max
Res	30291	0.691	0.052	0.619	0.844
Ressub1	47036	-1.445	0.968	-4.668	0.441
Ressub2	47069	32.538	22.855	1.320	97.620
Ressub3	30352	3.174	7.937	0.012	62.042
Ressub4	41419	0.104	0.182	-0.322	0.875
BioRisk	47107	0.045	0.050	0.000	0.100
Bioconcern1	47107	0.000	0.000	0.000	0.003
Bioconcern2	47107	0.000	0.000	0.000	0.002
Bioconcern3	47107	0.000	0.000	0.000	0.001
size	47107	22.072	1.286	19.777	26.113
TobinQ	47107	1.920	1.161	0.849	7.674
lev	47107	0.416	0.204	0.051	0.886
cash	47059	0.172	0.135	0.010	0.658
tangibility	47107	0.216	0.160	0.002	0.692
topten	47107	59.109	15.200	23.060	90.47

denote firm-specific characteristics. The model includes year, firm, and industry fixed effects to account for unobserved heterogeneity, where ϵ is the error term.

4. Results

4.1. Benchmark regression

Table 4 reports the regression results on the impact of biodiversity risk on SCR. The four models progressively incorporate additional fixed effects to improve robustness. Across all specifications, biodiversity risk shows a significant negative effect on SCR, with coefficients negative at the 1% significance level, indicating that higher biodiversity risk weakens firms' SCR.

4.2. Robustness

4.2.1. Substituting the key variables

To address potential estimation biases, this study conducts robustness checks. First, the dependent variable is redefined by (a) incorporating inventory changes into the resilience index and (b) recalculating recovery using total revenue and cost of goods sold (Table 5, Column 1–2). Second, the key explanatory variable is replaced with biodiversity risk concerns (Table 5, Columns 3–5).

Table 4
Benchmark regression.

	(1) Res	(2) Res	(3) Res	(4) Res
l.BioRisk	-0.021***	-0.017***	-0.018***	-0.020***
	(0.00)	(0.00)	(0.00)	(0.00)
size		-0.001***	-0.001*	-0.001
		(0.00)	(0.00)	(0.00)
lev		0.001	-0.000	-0.001
		(0.00)	(0.00)	(0.00)
topten		0.000***	0.000***	0.000***
		(0.00)	(0.00)	(0.00)
tangibility		-0.022***	-0.024***	-0.025***
		(0.00)	(0.00)	(0.00)
cash		0.026***	0.025***	0.023***
		(0.00)	(0.00)	(0.00)
TobinQ		0.001***	0.001***	0.001***
		(0.00)	(0.00)	(0.00)
Cons	0.690***	0.714***	0.697***	0.693***
	(0.00)	(0.01)	(0.01)	(0.01)
Firm FE	yes	yes	yes	yes
Year FE	yes	no	yes	yes
Industry FE	yes	no	no	yes
N	25492	28917	28917	25487
adj. R^2	0.757	0.751	0.751	0.761

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

4.2.2. Excluding special samples

To reduce potential distortions, observations from the COVID-19 period are excluded (Table 6, Column 1). Additionally, industries with fewer than 10 firms exposed to biodiversity risk are excluded to reduce the influence of extreme values (Column 2). Moreover, the five industries with the highest biodiversity risk identified by He et al. (2024) are sequentially excluded (Table 7). The results confirm the robustness of the baseline findings.

4.2.3. Considering provincial and city fixed effects

To control for unobserved regional factors such as local policies, economic conditions, and cultural differences, provincial and city fixed effects are added separately. The results (Table 6, Columns 3–4) indicate that the main findings remain consistent, further confirming the robustness of the results.

4.2.4. Addressing the issue of sample selection bias

To address potential selection bias, this study applies the Propensity Score Matching (PSM) method using firm size, leverage, ownership concentration, tangibility ratio, cash ratio, and Tobin's Q as covariates. A 1:1 kernel matching with replacement is performed. The results (Table 6, Column 5) indicate that the coefficient of biodiversity risk remains significantly negative, further supporting the robustness of the results.

4.2.5. Considering the impact of climate risk

To enhance robustness, this study incorporates climate risk variables, including provincial-level extreme low temperature days (LTD), high temperature days (HTD), rainfall days (ERD), and a comprehensive climate risk index as controls. These data are obtained from Guo et al. (2024). Table 8 shows that the coefficient of biodiversity risk remains significantly negative, consistent with the baseline regression.

4.2.6. IV regressions for endogeneity

To address endogeneity, this study uses a two-stage least squares (2SLS) approach, instrumenting with the interaction between corporate biodiversity risk and the ratio of nature reserve area to total jurisdictional area. The instrument appears to satisfy the exclusion restriction, as nature reserves focus on ecological conservation rather than firm operations. It captures exogenous regional variation and interacts with firm-level biodiversity risk, thereby mitigating potential direct effects on SCR. Table 9 indicates that the instrument significantly predicts biodiversity risk in the first stage, confirming its relevance and validity. In the second stage, the instrumented biodiversity risk has a significant negative effect on SCR, supporting the baseline findings.

4.3. Heterogeneity

To examine the heterogeneity, the sample is split by ownership, industry type, board gender diversity, and business diversification. Table 10 indicates a stronger negative effect in non-state-owned enterprises (non-SOEs), manufacturing firms, and firms with fewer female directors. Ownership heterogeneity may stem from private firms' weaker institutional pressure and resource constraints in adopting environmental practices (Hu & Chen, 2024). Notably, the negative impact of biodiversity risk is insignificant in SOEs, suggesting that organizational traits may buffer ecological risks. This may result from their greater stability and modernized operations (Ma et al., 2025). In addition, gender diversity may play a role, as firms with lower board gender diversity often struggle to advance environmental innovation under certain circumstances (Moreno-Ureba et al., 2022), weakening their ability to respond to biodiversity

Table 5
Robustness checks I.

	(1) Res2	(2) Res3	(3) Res	(4) Res	(5) Res
l.BioRisk	-0.018*** (0.00)	-0.018*** (0.00)			
l.Bioconcern1			-2.513*** (0.89)		
l.Bioconcern2				-3.706*** (1.27)	
l.Bioconcern3					-5.597** (2.49)
Cons	0.797*** (0.01)	0.797*** (0.01)	0.695*** (0.01)	0.695*** (0.01)	0.695*** (0.01)
Controls	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes	yes
N	25329	25318	25487	25487	25487
adj. R^2	0.764	0.764	0.761	0.761	0.761

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 6
Robustness checks II.

	(1) Model1	(2) Model2	(3) Model3	(4) Model4	(5) Model5
l.BioRisk	-0.019***	-0.020***	-0.020***	-0.020***	-0.020***
	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)
Cons	0.692***	0.692***	0.693***	0.693***	0.693***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Controls	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes	yes
N	19141	25388	25487	25329	25480
adj. R^2	0.737	0.761	0.761	0.762	0.761

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 7
Robustness checks III.

	(1) Res	(2) Res	(3) Res	(4) Res	(5) Res
l.BioRisk	-0.020***	-0.020***	-0.020***	-0.021***	-0.019***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)
Cons	0.696***	0.690***	0.691***	0.679***	0.690***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Controls	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes	yes
N	25126	25129	25279	24855	23682
adj. R^2	0.761	0.763	0.762	0.764	0.764

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 8
Robustness checks III.

	(1) Model1	(2) Model2	(3) Model3	(4) Model4
l.BioRisk	-0.020***	-0.020***	-0.020***	-0.020***
	(0.01)	(0.01)	(0.01)	(0.01)
LTD	-0.000			
	(0.00)			
HTD		0.000		
		(0.00)		
ERD			0.000	
			(0.00)	
ClimateRisk				0.000
				(0.00)
Cons	0.691***	0.690***	0.690***	0.689***
	(0.01)	(0.01)	(0.01)	(0.01)
Controls	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes
N	24195	24195	24195	24195
adj. R^2	0.760	0.760	0.760	0.760

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

risks.

Table 11 shows that the negative impact of biodiversity risk is more pronounced in less diversified firms. Diversification enhances supply chain agility through risk pooling and dual sourcing, enabling firms to manage uncertainty and sustain profitability under shocks (Liu et al., 2010; Lin et al., 2021). Regarding SCR sub-indicators, Table 12 shows that biodiversity risk adversely affects supply chain stability, growth, resistance, and recovery.

Table 9
Analysis for endogeneity concerns.

	1st stage BioRisk	2nd stage Res
l.BioRisk		-0.014**
		(0.006)
instru	0.095***	
	(0.000)	
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Industry FE	Yes	Yes
Obs.	25487	25487
adj. R^2		0.019
Anderson LM	1.6e+04	
p-value	0.000	
Cragg-Donald Wald	3.9e+04	

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 10
Heterogeneity analysis I.

	(1) SOEs	(2) NSOEs	(3) Non-mfg.	(4) Mfg.	(5) Fewer female dirs	(6) More female dirs
l.BioRisk	-0.007	-0.029***	0.000	-0.032***	-0.020***	-0.011
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Cons	0.703***	0.638***	0.766***	0.659***	0.689***	0.642***
	(0.02)	(0.01)	(0.02)	(0.01)	(0.02)	(0.02)
Controls	yes	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes	yes	yes
N	10423	14456	8872	16609	12813	12103
adj. R^2	0.784	0.765	0.757	0.771	0.780	0.785
p-value of the coefficient difference	0.000		0.000		0.000	

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

4.4. Mechanism

This section examines how biodiversity risk influences SCR. Inspired by Frank and Goyal (2003), a long-term financing gap is constructed to capture maturity mismatch (MM). Agency costs (AC) are proxied by the ratio of other receivables to total assets, following Luo et al. (2018). Table 13 shows that biodiversity risk significantly weakens SCR by exacerbating maturity mismatch. Table 14 indicates that biodiversity risk increases agency costs, leading to resource misallocation and weaker internal controls over biodiversity risk management.

Table 11
Heterogeneity analysis II.

	(1) Low divers	(2) High divers	(3) Low divers	(4) High divers
l.BioRisk	-0.021***	-0.017**	-0.022***	-0.018***
	(0.01)	(0.01)	(0.01)	(0.01)
Cons	0.672***	0.675***	0.677***	0.680***
	(0.02)	(0.02)	(0.02)	(0.02)
Controls	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes
N	11500	12184	11527	12167
adj. R^2	0.817	0.769	0.818	0.771
p-value of the coefficient difference	0.000		0.000	

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 12
Impact of Biodiversity Risks on SCR Sub-indicators.

	(1) Ressub1	(2) Ressub2	(3) Ressub3	(4) Ressub4
l.BioRisk	0.170**	-6.436***	2.590*	-0.101***
	(0.08)	(1.82)	(1.44)	(0.03)
Cons	-1.969***	91.702***	4.945	-2.004***
	(0.16)	(3.43)	(3.07)	(0.05)
Controls	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes
N	34470	34486	25539	34515
adj. R^2	0.732	0.766	0.173	0.222

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

5. Discussion

This study finds that biodiversity risk significantly undermines SCR, thereby enriching the theoretical connotations of the NRBV. In line with the NRBV's view that a firm's relationship with the natural environment constitutes a strategic determinant of competitive advantage (Hart, 1995), our findings highlight biodiversity as a distinct ecological constraint manifested through supply chains, underscoring the strategic importance of SCR for sustaining competitiveness under ecological pressures.

In terms of SCR measurement, unlike prior studies that focused on two capacity dimensions (Ma et al., 2025; Qi et al., 2024), this study adopts a broader framework encompassing four dimensions: resistance, stability, recovery, and growth. Findings show that biodiversity risk undermines the four sub-dimensions of SCR, thereby deepening our understanding of how ecological risks affect SCR.

Moreover, mechanism analyses identify agency costs as a key mediator, consistent with evidence that misaligned managerial incentives amplify vulnerability (Agha, 2013; Schramm et al., 2020). Maturity mismatch serves as another channel, echoing Yang and Li (2025), who document that firms exposed to high biodiversity risk often rely on short-term financing for long-term investments. These mechanisms suggest that resilience to biodiversity risk is not only an operational issue but also a financial and governance challenge.

Firm heterogeneity further shapes these effects. This paper finds that business diversification reduces vulnerability, offering new evidence to the ongoing debate on the role of diversification (Onali & Mascia, 2022; Buzhymska et al., 2024). In addition, board gender diversity mitigates biodiversity shocks, aligning with Gligor et al. (2024), who show that female leadership enhances supplier orientation. These findings underscore the critical role of organizational design in shaping firms' responses to biodiversity shocks.

6. Conclusion

This study shows that biodiversity risk significantly weakens SCR. Firms with limited diversification, fewer female directors, a manufacturing orientation, and non-state ownership are especially vulnerable. Mechanism analysis further reveals that maturity mismatch and agency costs are key channels.

It offers practical guidance: (1) Firms should integrate biodiversity indicators into risk management strategy, while policymakers can provide targeted incentives for diversification that enhance inclusiveness. (2) Financial instruments such as ecological bonds and insurance can help mitigate maturity mismatch, and governance mechanisms that reduce agency costs are critical in biodiversity-sensitive sectors. (3) Emerging economies should treat biodiversity loss amid rapid industrialization and urbanization as a critical risk and balance natural capital with climate governance to achieve sustainable development.

It should be noted that the SCR measurement in this study may not fully capture its complexity. Future research could improve

Table 13
Potential channel I.

	(1) MM	(2) Ressub1	(3) Ressub2	(4) Ressub3	(5) Ressub4	(6) Res
l.BioRisk	0.149***	0.142*	-6.458***	2.486*	-0.017	-0.015***
	(0.03)	(0.08)	(1.82)	(1.44)	(0.02)	(0.00)
MM		0.190***	0.149	0.664**	-0.556***	-0.031***
		(0.02)	(0.34)	(0.27)	(0.00)	(0.00)
Cons	2.366***	-2.419***	91.349***	3.446	-0.688***	0.763***
	(0.06)	(0.16)	(3.52)	(3.13)	(0.04)	(0.01)
Controls	yes	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes	yes	yes
N	34515	34470	34486	25539	34515	25487
adj. R^2	0.191	0.733	0.766	0.173	0.553	0.773

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 14
Potential channel II.

	(1) AC	(2) Ressub1	(3) Ressub2	(4) Ressub3	(5) Ressub4	(6) Res
l.BioRisk	0.006*	0.158*	-6.355***	2.541*	-0.095***	-0.019***
	(0.00)	(0.08)	(1.82)	(1.44)	(0.03)	(0.00)
AC		1.799***	-7.984***	4.762*	-0.850***	-0.058***
		(0.14)	(2.91)	(2.53)	(0.04)	(0.01)
Cons	0.067***	-2.086***	92.270***	4.527	-1.947***	0.699***
	(0.01)	(0.16)	(3.43)	(3.08)	(0.05)	(0.01)
Controls	yes	yes	yes	yes	yes	yes
Firm FE	yes	yes	yes	yes	yes	yes
Year FE	yes	yes	yes	yes	yes	yes
Industry FE	yes	yes	yes	yes	yes	yes
N	34501	34456	34472	25529	34501	25477
adj. R^2	0.420	0.733	0.766	0.173	0.233	0.761

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

resilience metrics by incorporating adaptability, flexibility, and forward-looking indicators. Finally, variations in China's ecological endowments and financial institutions underscore the importance of regional context in shaping SCR, highlighting the need to examine institutional and environmental heterogeneity to better inform global policy and practice.

Disclosure statement

None.

Funding

The work reported in the article is funded by National Social Science Fund of China (Study on the Goal Deviation and Mechanism Correction of the Financial inclusion Development based on the Theory of the Equality of Opportunity, 21BJY046).

CRediT authorship contribution statement

Wenwen Jin: Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Acknowledgements

The research presented in this study was supported by the National Social Science Fund of China (21BJY046). The author gratefully acknowledges the insightful reviews and comments from the editors and anonymous reviewers, that have significantly improved the manuscript. Any remaining errors are the sole responsibility of the author.

Data availability

Data are available upon reasonable request.

References

Ali, I., Arslan, A., Tarba, S., Mainela, T., 2023. Supply chain resilience to climate change inflicted extreme events in agri-food industry: The role of social capital and network complexity. Int. J. Prod. Econ. 264, 108968.
Andreoni, V., Miola, A., 2015. Climate change and supply-chain vulnerability: Methodologies for resilience and impacts quantification. Int. J. Emerg. Serv. 4 (1), 6–26.
Ahmad, M.F., Karpuz, A., 2024. Beyond climate change risk: Biodiversity and corporate cash holdings. Econ. Lett. 236, 111608.
Agha, M., 2013. Leverage, executive incentives and corporate governance. Account. & Finance 53 (1), 1–30.
Ahmed, Q., Sumbal, M.S., Lee, C., Tsui, E., 2025. Supply chain resilience and soft organizational factors–a bibliometric analysis and systematic literature review. Indust. Manage. Data Syst. 125 (4), 1360–1389.
Beck-O'Brien, M., Bringezu, S., 2021. Biodiversity monitoring in long-distance food supply chains: Tools, gaps and needs to meet business requirements and sustainability goals. Sustainability. 13 (15), 8536.
Bach, T.N., Hoang, K., Le, T., 2025. Biodiversity risk and firm performance: Evidence from US firms. Bus. Strategy. Environ. 34 (1), 1113–1132.
Brandon-Jones, E., Squire, B., Autry, C.W., Petersen, K.J., 2014. A contingent resource-based perspective of supply chain resilience and robustness. J. Supp. Chain Manage. 50 (3), 55–73.
Bulan, L.T., 2005. Real options, irreversible investment and firm uncertainty: new evidence from US firms. Rev. Financ. Econ. 14 (3-4), 255–279.
Buzhymska, K., Tsaruk, I., Biriuchenko, S., Pashchenko, O., Svitlyshyn, I., 2024. Impact of diversification on strategic business management. Sci. Bull. Mukachevo State Univ. Series "Econ. 11 (3), 34–46.
Carvalho, S.H.C.D., Cojoianu, T., Ascui, F., 2023. From impacts to dependencies: A first global assessment of corporate biodiversity risk exposure and responses. Bus. Strategy. Environ. 32 (5), 2600–2614.

Erol, I., Oztel, A., Dogru, T., Peker, I., Neuhofer, I.O., Benli, T., 2024. Supply chain resilience in the tourism and hospitality industry: A comprehensive examination of driving and restraining forces. Int. J. Hosp. Manage 122, 103851.
Frank, M.Z., Goyal, V.K., 2003. Testing the pecking order theory of capital structure. J. Financ. Econ. 67 (2), 217–248.
Fu, W., Feng, J., Gozgor, G., 2025. The Impact of Biodiversity Risk on Supply Chain Strategy: Evidence From the Chinese MNEs. Business Strategy and the Environment.
Ge, R., Bao, H., 2024. Digital transformation and resilience of supply chain in manufacture listed firms: A backward spillover effects in the vertical supply chain relationship. Financ. Res. Lett. 65, 105516.
Giglio, S., Kuchler, T., Stroebel, J., Zeng, X., 2023. Biodiversity risk (No. w31137). National bureau of economic research.
Gligor, D., Kashmiri, S., Golgeci, I., 2024. The Impact of Female Top Management Team Representation on a Firm's Supplier Orientation and Performance, 33. Production and operations management, pp. 2294–2310.
Ghadge, A., Wurtmann, H., Seuring, S., 2020. Managing climate change risks in global supply chains: a review and research agenda. Int. J. Prod. Res. 58 (1), 44–64.
Guo, K., Ji, Q., Zhang, D., 2024. A dataset to measure global climate physical risk. Data in Brief 54, 110502.
Gupta, S., Modgil, S., Choi, T.M., Kumar, A., Antony, J., 2023. Influences of artificial intelligence and blockchain technology on financial resilience of supply chains. Int. J. Prod. Econ. 261, 108868.
Hart, S.L., 1995. A natural-resource-based view of the firm, 20. Academy of management review, pp. 986–1014.
He, F., Chen, L., Lucey, B.M., 2024. Chinese corporate biodiversity exposure. Financ. Res. Lett. 70, 106275.
He, F., Duan, L., Lucey, B., Hao, J., 2025. Which factor affects corporate ESG rating divergence. Int. Rev. Financ. Anal. 104 (PA).
Hu, L., Chen, S., 2024. Ownership types and corporate environmental investment: Contingent effects of institutional and corporate factors. Int. Rev. Financ. Anal., 103893
Huang, Y., Créti, A., Jiang, B., Sanin, M.E., 2024. Biodiversity Risk, Firm Performance, and Market Mispricing. Firm Perform., Mark. Mispric. February 1, 2024.
Kedward, K., Ryan-Collins, J., Chenet, H., 2023. Biodiversity loss and climate change interactions: financial stability implications for central banks and financial supervisors. Clim. Policy. 23 (6), 763–781.
Lähtinen, K., Guan, Y., Li, N., Toppinen, A., 2016. Biodiversity and ecosystem services in supply chain management in the global forest industry. Ecosyst. Serv. 21, 130–140.
Li, Y., Liu, X., Canil, J., Cheong, C.S., 2025. Biodiversity risk and firm efficiency. Financ. Res. Lett. 71, 106414.
Li, Y., Zhang, Y., Geng, L., 2024. Digital finance, financing constraints and supply chain resilience. Int. Rev. Econ. Finance 96, 103545.
Lin, Y., Fan, D., Shi, X., Fu, M., 2021. The effects of supply chain diversification during the COVID-19 crisis: Evidence from Chinese manufacturers. Transp. Res. Part E: Logist. Transp. Rev. 155, 102493.
Liu, P., Liu, J., Tao, C., 2024. Market access, supply chain resilience and enterprise innovation. J. Innov. Knowl. 9 (4), 100576.
Liu, S., Lin, J., Hayes, K.A., 2010. An agile and diversified supply chain: reducing operational risks. Competitiveness review: An international business journal 20 (3), 222–234.
Liu, W., Chen, L., Luo, F., Zhao, Y., Li, X., Zeng, X., 2025. Novel assessment of China's cobalt supply chain resilience based on DPSIR model and machine learning. Resourc., Conserv. Recycl 215, 108107.
Lu, R., Jing, L., Yiliang, W., 2024. Party building ecosphere and supply chain resilience. J. Fin. Econ. 50 (07), 4C20.
Luo, J.H., Li, X., Chen, H., 2018. Annual report readability and corporate agency costs. China J. Account. Res. 11 (3), 187–212.
Lyu, Y., 2025. Government accounting supervision and corporate supply chain risk. Financ. Res. Lett. 74, 106785.
Ma, L., Wang, H., He, C., Sui, X., 2025. Can intelligent development improve manufacturing firms' supply chain resilience? Int. Rev. Econ. Finance 97, 103753.
Moreno-Ureba, E., Bravo-Urquiza, F., Reguera-Alvarado, N., 2022. An analysis of the influence of female directors on environmental innovation: When are women greener? J. Clean. Prod. 374, 133871.
Onali, E., Mascia, D.V., 2022. Corporate diversification and stock risk: Evidence from a global shock. J. Corpor. Finance 72, 102150.
Pandey, A.K., Daultani, Y., Pratap, S., 2024. Blockchain technology enabled critical success factors for supply chain resilience and sustainability. Bus. Strategy. Environ. 33 (2), 1533–1554.
Polasky, S., Daily, G., 2021. An introduction to the economics of natural capital. Rev. Environ. Econ. Policy. 15 (1), 87–94.
Qi, R., Ma, G., Liu, C., Zhang, Q., Wang, Q., 2024. Enterprise digital transformation and supply chain resilience. Financ. Res. Lett. 66, 105564.
Schramm, A., Schwemmer, A.H., Schymik, J., 2020. Capital (Mis) Alloc. Incen. Misalignm. *Available at SSRN 3704934*.
Su, Z., Deng, C., Feng, Y., Bai, Y., Xu, Z., 2025. Climate Risks and Corporate Supply Chain Configuration: Evidence from China. Res. Int. Bus. Finance, 102852.
Tennakoon, T.M.M.P., Chileshe, N., Rameezdeen, R., Ochoa, J.J., Samaraweera, A., 2025. Modelling the effect of transaction cost determinants and governance on Australian offsite construction supply chain resilience. Eng., Constr. Architect. Manage. 32 (13), 104–132.
Trinh, H.H., 2023. Biodiversity risks and corporate investment. *Available at SSRN 4587333*.
Türkeş, M.C., Stăncioiu, A.F., Marinescu, R.C., 2024. Modeling the impact of resilience factors and relational practice on performance of the supply chain. J. Innov. Knowl. 9 (3), 100533.
Wang, W., Chen, Y., Zhang, T., Deveci, M., Kadry, S., 2024b. The use of AI to uncover the supply chain dynamics of the primary sector: Building resilience in the food supply chain. Struct. Chang. Econ. Dyn. 70, 544–566.
Wang, T., Zou, T., Zhao, L., 2024a. Party organizations' participation in board governance and corporate supply chain stability. Financ. Res. Lett. 67, 105860.
Yang, G., Li, Y., 2025. A research on biodiversity risk and corporate short-term debt for long-term use under disclosure regulation. Financ. Res. Lett., 107049
Yin, L., Zhu, X., Li, J., 2025. Does exposure to biodiversity risk drive firms' digital transformation? Financ. Res. Lett. 82, 107489.
Zhong, J., Cheng, H., Jia, F., 2024. Supply chain resilience capability factors in agri-food supply chains. Oper. Manage. Res. 17 (3), 850–868.
Zhou, L., Tang, C., Cao, Y., 2025. Innovative human capital, government support for science and technology policy, and supply chain resilience. Financ. Res. Lett. 74, 106741.